

Davide Beltrame

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EDUCATION

Bocconi University

MSc in Artificial Intelligence (LM-18 Computer Science)

Milan, Italy

Sep 2024 – Jul 2026

- **Focus subjects:** Software Engineering, Algorithms for Optimisation and Inference, Deep Learning and Reinforcement Learning, Computer Vision, Complex Systems and Physical Models, Cryptography, Computational Neurosciences, NLP, Information Theory, Partial Differential Equations.
- **Research Thesis** ongoing in the Bocconi ML & Statistical Physics Lab on diffusion-based generation.
- **Student Associations:** BSML (see Experience); Author and Editor-in-Chief for STEM at **Consensus Rivista**

LUISS Guido Carli University

BSc in Management and Computer Science – final grade: 110/110 cum laude.

Rome, Italy

Sep 2021 – Jul 2024

- **Full scholarship recipient** (€33,000) as top 1 applicant.
- **Thesis:** *Machine Learning for Eco-climatic Strategy: Predicting Air Transport Emissions*. Supervisor: Giuseppe F. Italiano.
- **Student Associations:** President of the Starting Finance Club, Auditor at JELU Consulting

Liceo Scientifico "Augusto Righi"

High school diploma in scientific studies

Rome, Italy

Sep 2016 – Jun 2021

- **Final grade:** 100/100 with honours (con lode). Highest GPA (9.73/10, top 0.1%) in the institute.
- District-level (top 1%) in national Mathematics, Physics, Italian, Art History and Informatics Olympics.

EXPERIENCE

CommerceClarity

Machine Learning Engineer - Intern

Milan, Italy

May 2025 – Sep 2025

- Built a product image processor and compliance checker (format/quality/metadata) integrated in the ingestion pipeline; **reduced rejects from 5% to 0%**, unblocking listings and cutting manual rework.
- Proposed and delivered a company-wide analytics dashboard: added REST endpoints in the Laravel backend to expose growth, funnel and ops metrics; enabled monitoring of bottlenecks and weekly operating reviews.

Bocconi Students for Machine Learning (BSML)

Board Member & Researcher

Milan, Italy

Apr 2025 – Present

- Contributed to organizing tutorials and tech events, including a session on high-performance computing and a meetup with a Microsoft Azure AI Foundry engineer. Published project (2) below as part of BSML's research incubation.

LUISS Training Course for Italian Informatics Olympics

Tutor & Student Ambassador

Rome, Italy

Mar 2022 – May 2023

- Designed the **end-to-end contest pipeline**: authored problems, wrote reference solvers, built randomized input generators, curated test sets, and implemented auto-graders and score aggregation scripts.
- Selected among the top 4/70+ students; delivered algorithmic workshops (complexity, DP, graph algorithms).
- Founded the first Informatics Olympiad Discord community in Italy (**200+** students).

SELECTED RESEARCH PROJECTS

(1) More Efficient Text Diffusion via Length Prediction

Mar 2025

(2) Do AI and Human Experts See Pneumonia the Same Way?

May 2025

(3) (*MSc Thesis starting point*) Early-stopping Too Late? Traces of Memorization Before Overfitting in Generative Diffusion

Sep 2025 - ongoing

SKILLS & ADDITIONAL INFORMATION

Languages: Italian (native), English (fluent), French (intermediate), Spanish (basic), Japanese (elementary, Kana).

Programming: Python, C, R, SQL, C++, MATLAB, Linux. Scikit-Learn, Pandas, TensorFlow, Pytorch.

Tools: Git/GitHub, Parallel Computing in SLURM (HPC), Cloud basics, FastAPI, Docker.

Certifications: GRE General Test (QR 169, VR 163 - 92nd Percentile), IELTS 8.5/9 (C2), Google Analytics, KNIME.

Extracurricular: Kozminski University AI & NLP Program (Warsaw, 2023). MLI International Schools (Dublin, 2018). Haarlem (Netherlands) Model UN Delegate for Romania in the Economic and Social Council (2019). Mentee at LeadTheFuture (2025-ongoing).

Interests: Running, Cooking, Travelling, Cinema, Sci-fi literature, Art History. I took part in races in track & field and, more recently, road cycling. My view on life.